



Effect of different inter-repetition rest intervals across four load intensities on velocity loss and blood lactate concentration during full squat exercise

Ricardo Mora-Custodio, David Rodríguez-Rosell, Juan Manuel Yáñez-García, Miguel Sánchez-Moreno, Fernando Pareja-Blanco & Juan José González-Badillo

To cite this article: Ricardo Mora-Custodio, David Rodríguez-Rosell, Juan Manuel Yáñez-García, Miguel Sánchez-Moreno, Fernando Pareja-Blanco & Juan José González-Badillo (2018): Effect of different inter-repetition rest intervals across four load intensities on velocity loss and blood lactate concentration during full squat exercise, Journal of Sports Sciences, DOI: [10.1080/02640414.2018.1480052](https://doi.org/10.1080/02640414.2018.1480052)

To link to this article: <https://doi.org/10.1080/02640414.2018.1480052>



Published online: 30 May 2018.



Submit your article to this journal [↗](#)



View related articles [↗](#)



View Crossmark data [↗](#)



Effect of different inter-repetition rest intervals across four load intensities on velocity loss and blood lactate concentration during full squat exercise

Ricardo Mora-Custodio, David Rodríguez-Rosell, Juan Manuel Yáñez-García, Miguel Sánchez-Moreno, Fernando Pareja-Blanco and Juan José González-Badillo

Centro de Investigación en Rendimiento Físico y Deportivo, Universidad Pablo de Olavide, de Sevilla, España

ABSTRACT

This study aimed to analyze the acute effect of inter-repetition rest (IRR) intervals on mechanical and metabolic response during four resistance exercise protocols (REPs). Thirty resistance-trained men were randomly assigned to: continuous repetitions (CR), 10 s (IRR10) or 20 s (IRR20) inter-repetition rest. The REPs consisted of 3 sets of 6, 5, 4 and 3 repetitions against 60, 70, 75 and 80% 1RM, respectively, in the full squat exercise. Muscle fatigue was assessed using: percentage of velocity loss over three sets, percentage of velocity loss against the $\sim 1 \text{ m}\cdot\text{s}^{-1}$ load ($V1 \text{ m}\cdot\text{s}^{-1}$), and loss of countermovement jump (CMJ) height pre-post exercise. Blood lactate was measured before and after exercise. The percentage of velocity loss over three sets and lactate concentration were significantly lower ($P < 0.05$) for IRR groups compared to CR in all REPs. The CR group showed a significantly higher ($P < 0.05$) velocity loss against $V1 \text{ m}\cdot\text{s}^{-1}$ load and loss of CMJ height pre-post exercise than IRR groups in REP against 60% 1RM. In conclusion, both IRR groups produced a significant lower degree of fatigue compared to CR group. However, no significant differences were found in any measured variables between IRR configurations.

ARTICLE HISTORY

Accepted 15 May 2018

KEYWORDS

Resistance training; fatigue; metabolic stress; cluster configurations

Introduction

Muscle tissue structural and functional adaptations occur as a consequence of a particular blend of different exercise-related stressor factors including mechanical load, neuronal activation, hormonal adjustments and metabolic disturbances (Hoppeler, 2016). These stresses experienced are specific to the exercise training undertaken. Therefore, knowledge of the mechanical and physiological aspects underlying resistance training (RT) is essential to improve our understanding of the stimuli that affect adaptations (Crewther, Cronin, & Keogh, 2005; Sanchez-Medina & Gonzalez-Badillo, 2011). In this context, several studies have examined the effects of manipulating RT variables such as intensity (Aarskog, Wisnes, Wilhelmsen, Skogen, & Bjordal, 2012), number of repetitions performed in each set with respect to the maximum predicted number (García-Ramos et al., 2016; Gonzalez-Badillo et al., 2016; González-Hernández et al., 2017; Sanchez-Medina & Gonzalez-Badillo, 2011), number of sets (González-Hernández et al., 2017; Krieger, 2010; Tufano, Brown, & Haff, 2017), movement velocity (Pareja-Blanco, Rodríguez-Rosell, Sanchez-Medina, Gorostiaga, & Gonzalez-Badillo, 2014) and recovery time between sets (Rahimi, Qaderi, Faraji, & Boroujerdi, 2010). Another variable that is currently receiving great interest from researchers and practitioners is the introduction of inter-repetition rest (IRR) periods (i.e. a rest interval after every repetition or after a certain number of repetitions), traditionally known as cluster set (Haff et al., 2003; Tufano et al., 2017).

During a set performed with a traditional configuration (without IRR intervals), an involuntary decline in velocity, force

and power occurs as a consequence of fatigue accumulated during previous repetitions (Enoka & Duchateau, 2008; Izquierdo et al., 2006; Lawton, Cronin, & Lindsell, 2006). The continued use of this type of efforts during RT may generate undesired neuromuscular adaptations (Denton & Cronin, 2006; Lawton et al., 2006), as it has been reported that increase in strength as a consequence of training is velocity-specific (Englund, Sharp, Selsby, Ganesan, & Franke, 2017; Jones, Bishop, Hunter, & Fleisig, 2001; Pareja-Blanco et al., 2014, 2017). In accordance with this approach, the largest increases in strength occur at or near the training velocity, while a less effective training effect occurs if training velocity deviates from the target velocity (Englund et al., 2017; Jones et al., 2001; Pareja-Blanco et al., 2014). Therefore, training methods that allow maintenance of velocity, force and power output associated with every repetition during RT may optimize neuromuscular adaptation and maximize performance gains. In fact, a recent study (Pareja-Blanco et al., 2017) has reported that a lower degree of fatigue within the set resulted in greater improvements on the neuromuscular performance than a greater degree of fatigue. For this reason, several studies (Girman, Jones, Matthews, & Wood, 2014; Hardee, Triplett, Utter, Zwetsloot, & McBride, 2012; Iglesias-Soler et al., 2012) have suggested that introducing an IRR period could be an alternative method for attenuating the fatigue-induced performance decrements associated with traditional configuration in resistance exercise.

The current knowledge on cluster set indicates that this methodology allows greater force, velocity and power output

to be attained compared with traditional set configurations while reducing accumulated fatigue (Denton & Cronin, 2006; Garcia-Ramos et al., 2017; González-Hernández et al., 2017; Haff et al., 2003; Lawton et al., 2006; Moreno, Brown, Coburn, & Judelson, 2014). However, the response to different IRR periods remains relatively unexplored. It would be relevant to know the minimal IRR period that allows the maintenance of performance during different loading conditions. In this regard, Hardee et al. (2012) reported that adding IRR intervals (20 or 40 s) induced less decrements in peak velocity, force and power output over three sets of six repetitions during the power clean exercise compared to six successive repetitions. In addition, according with these authors, because there were no significant differences between 40 s and 20 s of IRR, it can be speculated that there is no additional benefit in resting more than 20 s (Hardee et al., 2012). More recently, other studies have examined the mechanical and metabolic responses to traditional and IRR set configurations in both the bench press (Garcia-Ramos et al., 2017) and the full squat (González-Hernández et al., 2017) exercises performed with the 10RM load and maintaining the same inter-set rest across the different IRR configurations. Taken together, these studies showed that IRR intervals of between 10–20 s induced similar degrees of fatigue, while IRR intervals of 30 s significantly reduced the velocity loss within the set and the metabolic stress (Garcia-Ramos et al., 2017; González-Hernández et al., 2017). However, these studies were conducted only using a relative intensity (10RM, ~75% 1RM) and completing repetitions to muscle failure. Considering that (a) this type of effort causes a marked disruption of cellular homeostasis, with considerable depletion of muscle purines and very large increases in lactate (14, 15); (b) increasing scientific evidence suggests that reaching repetition failure may not necessarily improve the magnitude of strength gains (Drinkwater et al., 2007; Izquierdo et al., 2006; Izquierdo-Gabarrén et al., 2010; Pareja-Blanco et al., 2017); and (c), most importantly, that ending each training set well ahead of failure seems to be a more appropriate stimulus to achieve improvements in specific athletic performance (Izquierdo et al., 2006; Izquierdo-Gabarrén et al., 2010; Pareja-Blanco et al., 2017), it seems necessary to conduct further studies with training protocols that are not performed until muscle failure.

RT is usually configured using multiple sets and different training loads (Crewther et al., 2005; González-Hernández et al., 2017). However, studies analyzing the acute effects of IRR intervals have used a single training set (Garcia-Ramos et al., 2015; Iglesias-Soler et al., 2012; Lawton et al., 2006; Moir, Graham, Davis, Guers, & Witmer, 2013) and/or a unique load magnitude (Garcia-Ramos et al., 2017; González-Hernández et al., 2017; Hansen, Cronin, & Newton, 2011; Hardee et al., 2012; Iglesias-Soler et al., 2012; Oliver et al., 2015; Rahimi et al., 2010). Therefore, it is difficult to extrapolate findings from this type of research to the responses associated with a typical RT session. Moreover, to the best of our knowledge, only four studies (Denton & Cronin, 2006; Garcia-Ramos et al., 2017; Girman et al., 2014; González-Hernández et al., 2017) have compared the acute metabolic response induced by IRR intervals with traditional set configurations. These studies (Denton & Cronin, 2006; Garcia-Ramos et al., 2017; Girman et al., 2014;

González-Hernández et al., 2017) concluded that cluster set configurations induced lower blood lactate concentrations than traditional sets, although results relating to protocols using the full squat exercise are scarce (González-Hernández et al., 2017).

In light of the aforementioned considerations, the aim of this study was to compare acute mechanical (losses of movement velocity and CMJ height) and metabolic (blood lactate concentration) responses to twelve different RT sessions, using four loading magnitudes (60, 70, 75 and 80% 1RM) and three IRR periods (0, 10 and 20 s) in resistance-trained men.

Methods

Participants

Thirty resistance-trained men (age: 22.8 ± 3.1 years; height: 1.80 ± 0.10 m; weight: 71.9 ± 10.9 kg) volunteered to participate in this study. Their estimated one-repetition maximum ($1RM_{est}$) strength (for more details, see testing procedure section) for the full squat exercise was 103.1 ± 12.7 kg (1.45 ± 0.24 normalized per kg of body mass). All participants were physically active sports science students with an RT experience ranging from 1.5 to 4 year (1–3 sessions per week) and were accustomed to performing the squat with correct technique. Participants were asked not to perform any strenuous RT during the course of the present study. After the first testing session, participants were matched according to their performance in $1RM_{est}$ in the full squat exercise and then randomly assigned to continuous repetitions (CR, $n = 10$), 10 s (IRR10, $n = 10$) or 20 s (IRR20, $n = 10$) interrepetition rest training protocols. After being informed about the purpose, testing procedures and potential risks of the investigation, participants gave their voluntary written consent to participate. No physical limitations or health problems were found after a medical examination. This study was conducted according to the Declaration of Helsinki and was approved by the research ethics committee of Pablo de Olavide University.

Experimental design

A descriptive design was used to analyze the acute mechanical and metabolic response evoked by twelve different resistance exercise protocols (REPs) performed with four loading conditions (60, 70, 75 and 80% 1RM) and three different IRR configurations: 0 s (CR), 10 s (IRR10) and 20 s (IRR20) in the full squat exercise. To address this, each group undertook 5 testing sessions separated by 7 days: 1) an initial test with increasing loads for the individual determination of $1RM_{est}$ strength and full load–velocity relationship in the full squat exercise (Sánchez-Medina, Pallarés, Pérez, Morán-Navarro, & González-Badillo, 2017); and 2) four different REPs differing in the rest between repetitions (0, 10 or 20 s). Thus, for each REP, all participants used the same number of sets (3 sets), repetitions per set (6, 5, 4 and 3 repetitions), loading magnitude (60, 70, 75 and 80% of 1RM, respectively), and inter-set rest (4 min). Obviously, since (i) the IRR interval was different between groups (0, 10 and 20s), and (ii) the inter-set rest was exactly the same (4 min) for all three experimental groups, the total resting time in the session and, consequently, the work-to-rest ratios were different between

groups. Thus, according to the different training configurations carried out in the present study, the CR group performed training sessions with the highest work-to-rest ratio and the IRR20 group had the lowest work-to-rest ratio. The REPs were configured as follows (set $\times$ repetitions [% 1RM]): 3×6 [60%] (REP 1); 3×5 [70%] (REP 2); 3×4 [75%] (REP 3); 3×3 [80%] (REP 4). The number of repetitions per set was selected based on the number of repetitions required to reach a velocity loss of $\sim 20\%$ within the set during traditional configurations, since a recent study has reported that this degree of fatigue produces greater improvements in neuromuscular performance than $\sim 40\%$ velocity loss within the set (near to muscle failure). The REPs were performed in random order and separated by 6–7 days. In order to compare the mechanical and metabolic responses following each REP, participants undertook a battery of measurements at pre-exercise (Pre) and post-exercise (Post): countermovement jump (CMJ) height; the individual load (kg) that elicits a $\sim 1.00 \text{ m}\cdot\text{s}^{-1}$ ($V1 \text{ m}\cdot\text{s}^{-1}$ load) mean propulsive velocity (MPV); and blood lactate concentration. During the course of the present study, participants did not perform any other RT. In addition, participants refrained from any strenuous physical activity for at least 2 days before each REP trial. All REPs were performed at the same time of day ($\pm 1 \text{ h}$) for each participant and under controlled environmental conditions (20°C and 55% humidity).

Testing procedures

Familiarization sessions and initial strength assessment

In the two weeks prior to the study, four familiarization sessions were carried out in order to ensure the use of proper technique in the full squat and CMJ exercises. These familiarization sessions consisted of 3 sets of 6, 5, 4 and 3 repetitions against loads of 60, 70, 75 and 80% 1RM, respectively, for the full squat exercise and five maximum CMJs in each session. The percentages of 1RM were estimated from the velocity-load relationship for this exercise (Sánchez-Medina et al., 2017). The first testing session was used to determine the participants' anthropometric data and to assess the individual load-velocity relationships and $1RM_{\text{est}}$ using a progressive isoinertial loading test in the full squat exercise. The warm-up and testing protocols were identical to those described elsewhere (Sánchez-Medina et al., 2017). The full squat was performed starting from the upright position with the knees and hips fully extended. Each participant descended in a continuous motion until the top of the thighs was below the horizontal plane, then immediately ascended back to the upright position. Participants were required to always execute the concentric phase of full squat at maximal intended velocity. This execution technique was carefully reproduced in all REPs performed in the study. Initial load was set at 30 kg and was progressively increased in 10 kg increments until the attained mean propulsive velocity (MPV) was $< 0.6 \text{ m}\cdot\text{s}^{-1}$ (which corresponds to $\sim 85\%$ 1RM) (Sánchez-Medina et al., 2017). Three repetitions were executed for light ($MPV \geq 1.10 \text{ m}\cdot\text{s}^{-1}$), two for medium ($1.10 \text{ m}\cdot\text{s}^{-1} > MPV \geq 0.80 \text{ m}\cdot\text{s}^{-1}$) and only one for heavy loads ($MPV < 0.80 \text{ m}\cdot\text{s}^{-1}$). Inter-set recoveries ranged from 3 min (light) to 5 min (heavy loads). Strong verbal encouragement was provided to motivate participants to give a maximal effort in each repetition. A Smith machine with no counterweight mechanism (Multipower Fitness Line, Peroga, Murcia, Spain) was used for all sessions. A dynamic

measurement system (T-Force System, Ergotech, Murcia, Spain) was used to measure movement bar velocity in every repetition during all testing sessions. The reliability of this system has been reported elsewhere (Sanchez-Medina & Gonzalez-Badillo, 2011). All reported repetition velocities in this study correspond to the MPV of the concentric phase (Sanchez-Medina, Perez, & Gonzalez-Badillo, 2010). The MPV was used in the present study for several reasons: 1) recent studies (Banyard, Nosaka, Vernon, & Haff, 2017; Garcia-Ramos, Pestana-Melero, Perez-Castilla, Rojas, & Haff, 2018) have shown that this variable has a very high intra- and inter-participant reliability, similar to mean velocity and peak velocity; 2) referring the mean values to only the propulsive phase (i.e. MPV) when assessing the velocity with which a load is lifted in a concentric action avoids underestimating an individual's neuromuscular ability, especially when lifting light and medium loads (Sanchez-Medina et al., 2010). In fact, it has been shown that when mean values are referred to the entire concentric portion of a lift (i.e. mean velocity), the higher the strength level (1RM) of a subject, the more underestimated his/her true strength potential gets when compared to that of another subject with a lower strength level (Sanchez-Medina et al., 2010). In other words, the MPV allows better discrimination between subjects with different performance levels compared to the mean velocity and peak velocity. For these reasons, the assessment of strength performance in a single trial and the analysis of the training effect should preferentially be done by referring mean values to the propulsive phase (MPV), especially when using loads $\leq 70\%$ 1RM (Sanchez-Medina et al., 2010).

Resistance exercise protocols

The characteristics of each REP are displayed in Table 1. The 12 types of REP were performed always using three sets and 4-min inter-set recovery periods. The number of repetitions per set was 6 (60% 1RM), 5 (70% 1RM), 4 (75% 1RM) and 3 (80% 1RM), respectively. The load magnitude for each participant in each REP was determined from the load-velocity relationship for the full squat exercise, as it has recently been shown that there is a very close relationship between percentage of 1RM and MPV (Sánchez-Medina et al., 2017). Thus, the target MPV to be attained in the first (usually the fastest) repetition of the first exercise set in each REP was used as an estimation of percentage of 1RM, as follows: $\sim 0.98 \text{ m}\cdot\text{s}^{-1}$ (60% 1RM), $\sim 0.82 \text{ m}\cdot\text{s}^{-1}$ (70% 1RM), $\sim 0.75 \text{ m}\cdot\text{s}^{-1}$ (75% 1RM), and $\sim 0.68 \text{ m}\cdot\text{s}^{-1}$ (80% 1RM) (Sánchez-Medina et al., 2017). Consequently, before starting each REP, adjustments in the proposed load (kg) were made when needed so that the velocity of the first repetition matched the programmed velocity ($\pm 0.03 \text{ m}\cdot\text{s}^{-1}$).

In each testing session the warm-up consisted of 5 min of jogging at a self-selected easy pace, 5 min of joint mobilization exercises, two 30-m running accelerations, two sets of 10 squats with no external load (own body weight) and five CMJs with increasing intensity. Then, three maximal CMJs, separated by 20 s rests, were performed and the mean jump height (cm) was taken as the pre-exercise reference value. The CMJ height was determined using an infrared timing system (Optojump, Microgate, Bolzano, Italy). This was followed by the determination of the $V1 \text{ m}\cdot\text{s}^{-1}$ load. For this purpose, participants performed three sets of six down to three repetitions (3-min rests)

Table 1. Descriptive characteristics of the resistance exercise protocol performed in each session.

Scheduled	REP 1	REP 2	REP 3	REP 4
Sets x rep	3 x 6	3 x 5	3 x 4	3 x 3
Target MPV ($\text{m}\cdot\text{s}^{-1}$)	0.98 (~60% 1RM)	0.83 (~70% 1RM)	0.75 (~75% 1RM)	0.68 (~80% 1RM)
Actually Performed				
Reference MPV ($\text{m}\cdot\text{s}^{-1}$)				
CR	1.00 ± 0.02 (~58.8% 1RM)	0.84 ± 0.02 (~69.3% 1RM)	0.75 ± 0.03 (~75.1% 1RM)	0.69 ± 0.02 (~79.0% 1RM)
IRR10	1.00 ± 0.03 (~58.8% 1RM)	0.83 ± 0.03 (~69.9% 1RM)	0.76 ± 0.02 (~74.5% 1RM)	0.68 ± 0.02 (~79.6% 1RM)
IRR20	0.99 ± 0.03 (~59.4% 1RM)	0.84 ± 0.02 (~69.3% 1RM)	0.75 ± 0.02 (~75.1% 1RM)	0.68 ± 0.01 (~79.6% 1RM)
Average MPV over three sets ($\text{m}\cdot\text{s}^{-1}$)				
CR	0.88 ± 0.05	0.71 ± 0.05	0.64 ± 0.05	0.59 ± 0.05
IRR10	0.92 ± 0.04 *	0.73 ± 0.04	0.68 ± 0.04	0.60 ± 0.04
IRR20	0.92 ± 0.05 *	0.78 ± 0.03 **	0.69 ± 0.03 *	0.63 ± 0.02 *

REP: Resistance exercise protocol; MPV: Mean propulsive velocity; CR: continuous repetition group; IRR10: 10 seconds interrepetition rest group; IRR20: 20 seconds interrepetition rest group; 1RM: One-repetition maximum. Statistically significant differences with respect to CR: * $P < .05$, ** $P < .01$.

with increasing loads (from 20 kg) up to $V1 \text{ m}\cdot\text{s}^{-1}$ load ($\pm 0.03 \text{ m}\cdot\text{s}^{-1}$). This value was chosen because it is a sufficiently high velocity, which is attained against a medium load (~60% RM in the SQ), and it allows a good expression of the effect of loading on velocity, besides being a relatively easy-to-move and well tolerated load (Sanchez-Medina & Gonzalez-Badillo, 2011). The $V1 \text{ m}\cdot\text{s}^{-1}$ load (kg) was thus taken as a pre-exercise reference measure against which to compare the velocity loss experienced after the three exercise sets (Sanchez-Medina & Gonzalez-Badillo, 2011). Finally, after load adjustments in the proposed load (kg), the three sets of the corresponding REP were performed next. In those REPs with IRR intervals the barbell rested on the supports of the Smith machine during the rest periods. Immediately after completing the last repetition of the third training set, participants performed three maximal CMJs, separated by 20-s rests. In addition to CMJs, the participants also executed three maximal-effort consecutive repetitions against the $V1 \text{ m}\cdot\text{s}^{-1}$ load (load was changed within 10–15 s with the help of trained spotters). The CMJ and $V1 \text{ m}\cdot\text{s}^{-1}$ load mean values were taken as the immediate post-exercise measures. After 1 min rest, blood samples were taken for the analysis of lactate.

Measurements of fatigue

Three different methods were used to quantify the extent of fatigue induced by each REP (Sanchez-Medina & Gonzalez-Badillo, 2011). The first method analyzed the decline in repetition velocity during the three consecutive exercise sets and was calculated as the percentage loss in MPV from the fastest to the slowest repetition of each set and averaged over the 3 sets. The second method examined the pre-post exercise change in velocity attained against the $V1 \text{ m}\cdot\text{s}^{-1}$ load. The third method involved the calculation of percentage change in CMJ height after the three exercise sets. It was calculated as the percentage change in CMJ height from pre- to post-exercise (see Figure 1).

Blood lactate concentration

Capillary whole blood samples were drawn from the earlobe before exercise and again 1 min after completing the last

repetition of the $V1 \text{ m}\cdot\text{s}^{-1}$ load in each REP. The Lactate Pro 2 LT-1730 (Arkray, Kyoto, Japan) portable lactate analyzer was used for lactate measurements. The suitability and reproducibility of this analyzer has been previously established throughout the physiological range of $1.0\text{--}18.0 \text{ mmol}\cdot\text{L}^{-1}$ (Pyne, Boston, Martin, & Logan, 2000).

Statistical analysis

Values are reported as mean $\pm$ standard deviation (SD). Homogeneity of variance across groups (CR vs. IRR10 vs. IRR20) was verified using Levene's test, whereas the normality of distribution of the data was examined with the Kolmogorov-Smirnov test. A one-way analysis of variance (ANOVA) was conducted on the MPV attained in the first repetition of each set configuration, the CMJ height reached at the beginning of each testing session, and blood lactate concentration, in order to examine possible differences in these variables between the different set configurations (CR vs. IRR10 vs. IRR20) against each training load. A two-way factorial ANOVA with an inter-subject factor (Group: CR, IRR10, IRR20) and a repeated measures factor (Repetitions: 6, 5, 4, 3) was used to compare MPV of each repetition in each REP with Bonferroni post-hoc comparisons to clarify the interaction. A one-way ANOVA was used to examine the differences between training configurations (CR, IRR10, IRR20) in mechanical and metabolic responses in each REP. Bonferroni post-hoc comparisons were used to analyze the differences between experimental groups. Intra-group effect sizes (ES, 95% confidence interval) were calculated using Hedge's g (Hedges & Olkin, 1985). Threshold values for assessing magnitudes of standardized effects (changes as a fraction or multiple of baseline SD) were 0.20, 0.60, 1.20, and 2.00 for small, moderate, large, and very large, respectively (Hopkins, Marshall, Batterham, & Hanin, 2009). Statistical significance was established at $P < 0.05$. Statistical analyses were performed using SPSS version 18.0 (SPSS Inc., Chicago, IL, USA).

Results

All variables were normally distributed and homoscedastic across REPs. No significant differences between groups were found prior to the experimental protocols for any of the variables analyzed.

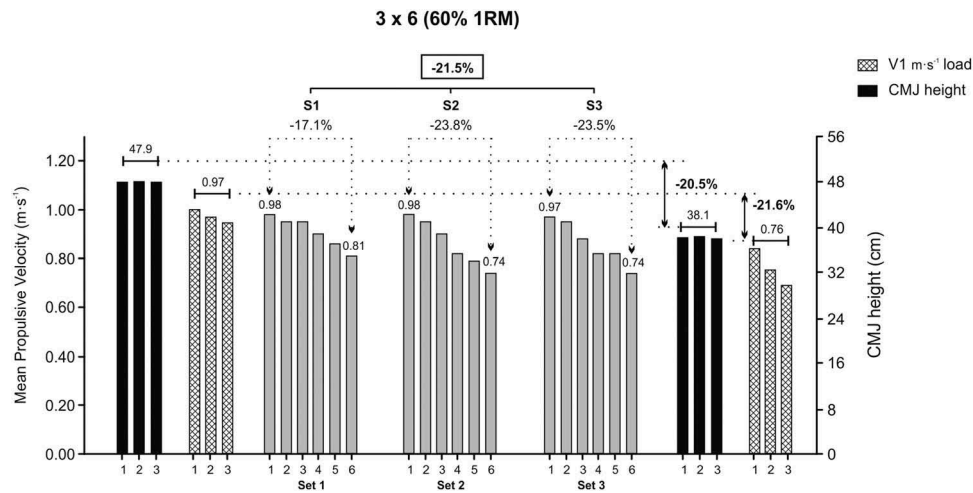


Figure 1. Example of performing resistance exercise protocol 1 for a representative participant of CR. Relative MPV loss over three sets (21.5%), MPV pre-post exercise against the V1 m·s⁻¹ load (21.6%) and loss of CMJ height pre-post exercise (20.5%) were calculated.

Resistance exercise protocols characteristics

Averages of the best MPV and averages of the MPV over the three sets for each REP in each group are reported in Table 1. No significant differences between-groups were found in the best MPV in any REP. The average MPV over the three sets was significantly greater for IRR20 compared to CR in all REPs, whereas differences between IRR10 and CR only were significant for REP 1 (Table 1). No significant differences between IRR10 and IRR20 were observed in the average MPV over the three sets in any of the four REPs performed.

There was significant 'group x repetition' interaction ($P < 0.001$) in the velocity decline pattern within each set configuration for all the four REPs under study. Bonferroni post-hoc comparisons revealed significant differences between CR and IRR10 ($P < 0.05 - 0.001$) and CR and IRR20 ($P < 0.05 - 0.001$) in MPV of each repetition, mainly in the last two repetitions of each training set

(Figure 2). No significant differences for MPV were observed between both IRR configurations. The intra-group analysis of velocity decline pattern within each set configuration (Figure 2) showed that the MPV significantly decreased ($P < 0.05 - 0.01$) from the second repetition in all three sets of any REP for CR. For IRR10, significant decrements ($P < 0.05 - 0.01$) in MPV were found from the second (REP 4) and third (REP 1, 2 and 3) repetition of each set, while for IRR20 the MPV declines were statistically significant ($P < 0.05$) from the third (REP 4), fourth (REP 3) and fifth (REP 1 and 2) repetition of each set.

Mechanical and metabolic measurements of fatigue

Comparison between groups showed that the percentage of MPV loss over the three sets and post-exercise blood lactate concentration were significantly ($P < 0.05 - 0.001$) greater for

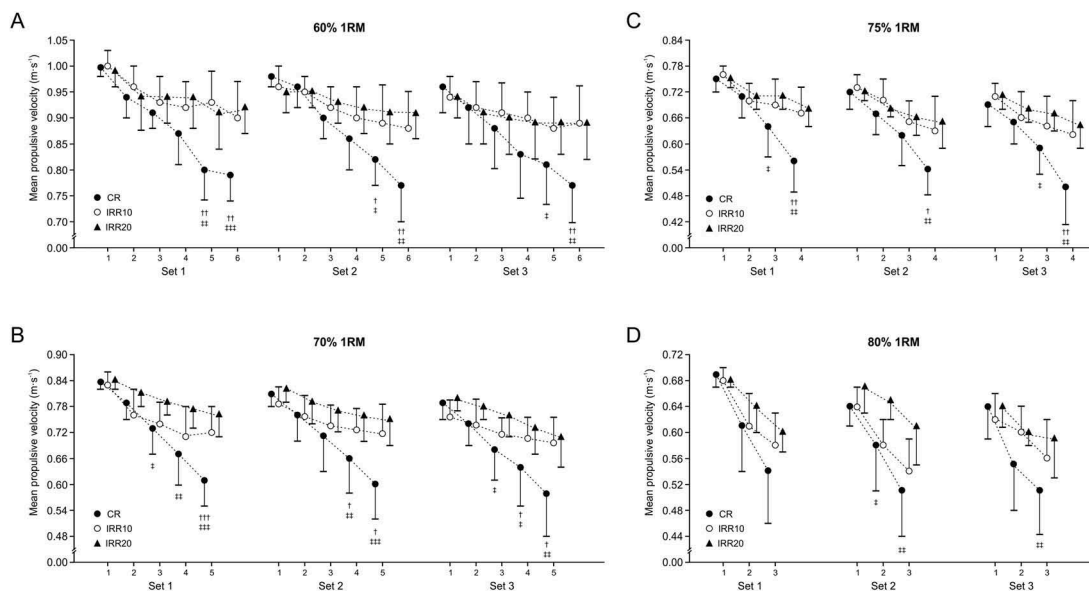


Figure 2. Average mean propulsive velocity during each repetition for each load magnitude used (60%, 70%, 75% and 80% 1RM) in the different set configurations (CR, IRR10 and IRR20). Statistically significant differences respect to IRR10: † $P < .05$, †† $P < .01$, ††† $P < 0.001$. Statistically significant differences respect to IRR20: ‡ $P < .05$, ‡‡ $P < .01$, ‡‡‡ $P < .001$.

Table 2. Mechanical and metabolic measurements of fatigue after each effort for each group.

Training Configuration	Loss of MPV over three sets (%)	Lactate (mmol·L ⁻¹)	Loss of CMJ		Loss of MPV with V1 m·s ⁻¹ Load	
			Δ (%)	ES (95% IC)	Δ (%)	ES (95% IC)
REP 1 (3x6 at 60% 1RM)						
CR	22.1 ± 5.7 ***†††	6.7 ± 2.9 ***†††	21.3 ± 5.3 *	1.56 (0.71 – 2.41)	22.8 ± 8.8 *†	2.94 (1.18 – 4.71)
IRR10	11.2 ± 5.4	3.4 ± 0.9	17.2 ± 4.5	1.38 (0.50 – 2.26)	13.6 ± 8.1	1.93 (0.76 – 3.10)
IRR20	9.1 ± 3.3	2.9 ± 0.8	13.6 ± 5.0	0.86 (0.32 – 1.41)	12.6 ± 8.5	1.89 (0.67 – 3.12)
REP 2 (3x5 at 70% 1RM)						
CR	27.4 ± 8.6 ***†††	6.6 ± 2.7 ***††	21.5 ± 4.7	1.62 (0.75 – 2.50)	21.2 ± 10.1	2.35 (0.92 – 3.78)
IRR10	13.3 ± 4.8	3.7 ± 0.9	16.4 ± 6.0	1.44 (0.65 – 2.24)	12.8 ± 7.0	1.86 (0.85 – 2.87)
IRR20	12.1 ± 4.3	2.9 ± 0.7	18.1 ± 4.1	1.34 (0.66 – 2.03)	13.5 ± 9.0	2.16 (0.97 – 3.35)
REP 3 (3x4 at 75% 1RM)						
CR	26.9 ± 9.0 ***†††	5.5 ± 2.1 ***†	19.5 ± 5.2	1.23 (0.57 – 1.89)	22.1 ± 11.3	1.70 (0.49 – 2.92)
IRR10	14.1 ± 5.9	3.9 ± 1.2	14.6 ± 5.1	1.08 (0.50 – 1.65)	15.7 ± 7.5	2.01 (0.97 – 3.04)
IRR20	11.3 ± 3.3	2.9 ± 0.9	17.0 ± 3.6	1.17 (0.53 – 1.81)	15.8 ± 6.9	2.16 (1.06 – 3.26)
REP 4 (3x3 at 80% 1RM)						
CR	22.1 ± 8.8**†	4.9 ± 1.6 ***	18.2 ± 5.8	1.05 (0.45 – 1.65)	16.8 ± 9.7	1.96 (0.77 – 3.14)
IRR10	13.2 ± 5.6	3.6 ± 1.2	16.6 ± 4.8	1.08 (0.41 – 1.75)	18.4 ± 7.7	2.47 (1.19 – 3.76)
IRR20	12.0 ± 5.3	2.6 ± 0.5	16.2 ± 3.8	0.91 (0.44 – 1.38)	16.7 ± 7.9	2.64 (1.19 – 4.09)

CR: Continuous repetition group; IRR10: 10 seconds interrepetition rest group; IRR20: 20 seconds interrepetition rest group; statistically significant differences respect to IRR20: * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$; statistically significant differences respect to IRR10: † $P < 0.05$, †† $P < 0.01$, ††† $P < 0.001$.

CR in all REPs compared to IRR10 and IRR20, whereas no significant differences between IRR10 and IRR20 were found (Table 2). The CR group showed significantly ($P < 0.01$) greater loss of CMJ height than IRR20 in REP 1, and significantly higher loss of MPV against V1 m·s⁻¹ load than IRR10 ($P < 0.05$) and IRR20 ($P < 0.01$) in REP 1. For REP 2, REP 3 and REP 4, although there were no significant differences between groups, CR showed a greater percentage of change (16.8 – 22.1%) and ES (1.05 – 2.35) compared to IRR10 (Δ: 12.8 – 18.4%; ES: 1.08 – 2.47) and IRR20 (Δ: 13.5 – 18.1%; ES: 0.91 – 2.64) in these variables (Table 2). No statistically significant or practical differences were observed in loss of CMJ and MPV against V1 m·s⁻¹ load between IRR10 and IRR20 for any REP (Table 2).

Discussion

The aim of our study was to compare mechanical and metabolic responses to four REPs performed with three different IRR periods (0, 10 and 20 s) during full squat exercise in resistance-trained men. As expected, CR presented a significant greater velocity loss within the set and a higher lactate concentration compared to both IRR10 and IRR20 in all REPs analyzed. Despite this, loss of CMJ height and loss of MPV against the V1 m·s⁻¹ load was significantly higher for CR compared to IRR10 and IRR20 only in the REP performed against 60% 1RM. In contrast, no significant differences were observed between IRR10 and IRR20 configurations in any measured variables. Therefore, the results of the present study confirm that for a training session with the same volume and relative load, introducing a recovery time between repetitions reduces the degree of induced fatigue during RT. However, despite these findings, further studies need to be conducted in order to determine whether RT programs with IRR configurations may result in greater improvements in neuromuscular performance than CR training programs.

A unique and important aspect of this investigation was that, by monitoring repetition velocity and adjusting the actual loads to be lifted from the load-velocity relationship for the full squat

exercise (Sánchez-Medina et al., 2017), we made sure that all participants in each experimental group used a very similar relative load in each REP (Table 1). Because the movement velocity and power output decline, and consequently induced fatigue, are influenced by the relative load (% 1RM) used (Izquierdo et al., 2006), it appears essential in studies analyzing the acute effect of different RT configurations that all participants are tested with the same relative load in order to truly understand the impact of varying the IRR interval. However, previous studies in this area (García-Ramos et al., 2015, 2017; Girman et al., 2014; González-Hernández et al., 2017; Haff et al., 2003; Hansen et al., 2011; Hardee et al., 2012; Lawton et al., 2006; Moir et al., 2013; Oliver et al., 2015) did not use the load-velocity relationship (the fastest MPV or mean velocity of the set) to determine loading magnitude (Sánchez-Medina et al., 2017). Protocols used in most previous studies were prescribed according to the maximal load (kg) that could be lifted in a determined number of maximum repetitions (e.g. 4, 6, 8, 10 and 12RM) (García-Ramos et al., 2017; González-Hernández et al., 2017; Lawton et al., 2006) or as a given percentage of 1RM measured in an initial session (García-Ramos et al., 2015; Girman et al., 2014; Oliver et al., 2015), i.e. these studies did not check whether the actual relative load used for participants corresponded with the proposed load (Gonzalez-Badillo & Sanchez-Medina, 2010; Gonzalez-Badillo, Yanez-Garcia, Mora-Custodio, & Rodriguez-Rosell, 2017; Sánchez-Medina et al., 2017). Therefore, to the best of our knowledge, this is the first study analyzing the acute effect of different IRR configurations compared to CR configurations where the actual relative load used for all participants was equated across the load-velocity relationship in the full squat exercise.

According to our results, most studies (García-Ramos et al., 2015, 2017; González-Hernández et al., 2017; Haff et al., 2003; Hansen et al., 2011; Hardee et al., 2012; Lawton et al., 2006; Oliver et al., 2015) analyzing the effect of adding a rest interval after every repetition on changes in different kinetic and kinematic variables have reported significantly lower velocity and power output repetition declines across the set for IRR compared to CR configurations. These results are consistent in

a variety of resistance exercises, including squat (Oliver et al., 2015), bench press (Garcia-Ramos et al., 2015, 2017; Lawton et al., 2006), power clean (Haff et al., 2003; Hansen et al., 2011), and jump squat (Hardee et al., 2012), and indicate that the inclusion of IRR periods may allow an increase in the number of repetitions performed at greater velocities. In connection with the loss of MPV over the set, CR also showed greater intra-group ES and percentages of losses of MPV against $V1 \text{ m}\cdot\text{s}^{-1}$ load and CMJ height (mainly in REP1), and higher blood lactate concentrations (in all REPs) compared to both IRR10 and IRR20 (Table 2). Losses in CMJ height and velocity attained against $V1 \text{ m}\cdot\text{s}^{-1}$ load, along with lactate concentration, have been considered as reliable and effective methods to determine the acute and delayed (72 h post-effort) degree of fatigue produced by different RT protocols using CR (Gonzalez-Badillo et al., 2016; Sanchez-Medina & Gonzalez-Badillo, 2011). However, only a few studies (Denton & Cronin, 2006; Garcia-Ramos et al., 2017; Girman et al., 2014; González-Hernández et al., 2017; Oliver et al., 2015) have analyzed the degree of fatigue or metabolic stress after RT with different IRR configurations. In line with our results, these studies using the squat exercise reported that RT protocols conducted with IRR configurations showed significantly lower losses of CMJ height (Girman et al., 2014; González-Hernández et al., 2017) and blood lactate concentrations (Denton & Cronin, 2006; Girman et al., 2014; González-Hernández et al., 2017; Oliver et al., 2015) in comparison with CR configurations.

It appears that velocity and power performance are primarily dependent upon the phosphagen system (Girman et al., 2014). It has been observed that the phosphocreatine (PCr) stores progressively diminish as the number of repetitions increases, reaching near-complete depletion when performing a single set to muscle failure using a CR configuration (Gorostiaga et al., 2012). Therefore, it appears that the inclusion of short IRR periods would provide a partial replenishment of PCr, which is important for velocity and power output maintenance during repeated bouts of exercise (Bogdanis, Nevill, Boobis, Lakomy, & Nevill, 1995). In this regard, previous studies have speculated that at least a 15–20 s IRR interval between explosive exercises may be needed for PCr resynthesis (Harris et al., 1976) and maintenance of blood lactate concentration and physical performance (Hardee et al., 2012; Pereira et al., 2008). However, in contrast with this hypothesis, our results suggest that lower IRR intervals (10 s) could be sufficient to avoid important velocity declines during and after multiple sets and repetitions during a resistance training session, at least in training protocols where less than half the maximum possible number of repetitions were completed in each training set (Table 2 and Figure 2). However, our results do not allow us to know whether when performing a greater number of repetitions per set, resting 10 or 20 s between repetitions induces different or similar degree of fatigue, and further studies are needed to resolve this matter. In addition, a recent study (González-Hernández et al., 2017) using a similar methodology to that used in the present study showed similar loss of CMJ height and lactate concentration between 10 s and 15 s of IRR intervals. However, when the recovery time between repetitions lasted until 30 s, both loss of CMJ and lactate concentration were significantly lower compared to

the 10 and 15 s IRR intervals (González-Hernández et al., 2017). Taken together, these results appear to indicate that, for the same RT protocol, an IRR interval range from 10–20 s induces a similar degree of fatigue, whereas lower performance declines are found when IRR intervals of 30 s are allowed.

Conclusions

The results of the present study suggest that adding an IRR period of 10 s or 20 s results in a significantly lower (i) velocity loss during and after the set; (ii) loss of CMJ height; and (iii) blood lactate concentrations compared to CR configurations in a wide range of loading intensities (60%, 70%, 75% and 80% 1RM). However, no significant differences were observed in these variables between 10 s and 20 s IRR intervals. Therefore, it appears that the inclusion of 10 s IRR configuration alleviated fatigue-induced decreases in velocity, strength and jump height at a very similar level to 20 s IRR periods, at least in REPs similar to those performed in the present study. In addition, 10 s IRR requires a lower work-to-rest ratio than 20 s IRR. Despite these findings, further longitudinal studies are needed to determine whether the greater movement velocity in each repetition produced by IRR configurations results in greater improvements in neuromuscular performance compared to CR configurations.

Practical applications

The main practical application of the present study was that both IRR configurations presented a significant lower degree of fatigue than CR configuration. This may be very important for coaches and physical fitness professionals because they could apply this type of RT configurations for maximizing velocity, power output and force attained in each repetition during a training set. In addition, no significant differences were observed between IRR10 and IRR20. Therefore, the use of IRR10 would induce a very similar mechanical stimulus to IRR20. However, it is important to also note that a 10 s IRR configuration requires a shorter total training time than a 20 s of IRR interval (i.e. it is more efficient). This is particularly relevant for many athletes whose goals are focused on maximizing performance in the most efficient way. Therefore, coaches and physical conditioning professionals should consider using 10 or 20 s of IRR intervals during RT in order to maintain a greater movement velocity in each repetition and a lower degree of fatigue during a multiple set exercise protocol against different relative loads.

Limitations

The present study has some limitations that need to be addressed. Obviously, one of the main limitations of this study is the impossibility of testing the IRR as a unique independent variable. This problem is inherent to any IRR configurations. Including a rest interval between repetitions or group of repetitions directly compromises the recovery time between sets, total resting time or work-to-rest ratio. In this regard, if the total resting time is maintained across different cluster or IRR configurations, the inter-set rest interval will be

different. Conversely, if the inter-set recoveries are the same for the different training programs used, the total resting time will be greater and, consequently, the work-to-rest-ratio will be lower in those training programs with longer recovery times between repetitions. This factor directly influences on the mechanical and metabolic responses to RT, so that the group with the highest work-to-rest ratio (e.g. CR) will tend to show a greater degree of fatigue and metabolic stress. Conversely, the group with the smallest work-to-rest ratio (e.g. IRR20), will show a higher repetition velocity in each training set, and lower fatigue and metabolic stress, similar to what was shown in the present study. Therefore, it appears that is not possible to design training programs differing only in the IRR interval. In this situation, the aim should be to analyze the acute and chronic effects of different IRR configurations equaling recovery times between sets or total resting time (i.e. work-to-rest ratio) in order to know what RT configuration is the most efficient. The present study, along with other recently published studies (Garcia-Ramos et al., 2017; González-Hernández et al., 2017) using a similar methodology, could constitute a first step in acquiring this knowledge. Another possible limitation is the low number of participants in each group. Thus, some effects are associated with large confidence limits for the intra- and inter-group change differences. Therefore, we cannot be sure whether differences within and between groups would have been clearer with a larger number of subjects in each experimental group.

Acknowledgments

The authors declare no conflicts of interest

Disclosure statement

No potential conflict of interest was reported by the authors.

References

- Aarskog, R., Wisnes, A., Wilhelmsen, K., Skogen, A., & Bjordal, J. M. (2012). Comparison of two resistance training protocols, 6RM versus 12RM, to increase the 1RM in healthy young adults. A single-blind, randomized controlled trial. *Physiotherapy Research International: the Journal for Researchers and Clinicians in Physical Therapy*, 17, 179–186.
- Banyard, H. G., Nosaka, K., Vernon, A. D., & Haff, G. G. (2017). The reliability of individualized load-velocity profiles. *International Journal of Sports Physiology and Performance*, 1–22. doi:10.1123/ijsp.2017-0610
- Bogdanis, G. C., Nevill, M. E., Boobis, L. H., Lakomy, H. K., & Nevill, A. M. (1995). Recovery of power output and muscle metabolites following 30 s of maximal sprint cycling in man. *The Journal of Physiology*, 482(Pt 2), 467–480.
- Crewther, B., Cronin, J., & Keogh, J. (2005). Possible stimuli for strength and power adaptation: Acute mechanical responses. *Sports Medicine (Auckland, N.Z.)*, 35, 967–989.
- Denton, J., & Cronin, J. B. (2006). Kinematic, kinetic, and blood lactate profiles of continuous and intraset rest loading schemes. *Journal of Strength and Conditioning Research / National Strength & Conditioning Association*, 20, 528–534.
- Drinkwater, E. J., Lawton, T. W., McKenna, M. J., Lindsell, R. P., Hunt, P. H., & Pyne, D. B. (2007). Increased number of forced repetitions does not enhance strength development with resistance training. *Journal of Strength and Conditioning Research / National Strength & Conditioning Association*, 21, 841–847.
- Englund, D. A., Sharp, R. L., Selsby, J. T., Ganesan, S. S., & Franke, W. D. (2017). Resistance training performed at distinct angular velocities elicits velocity-specific alterations in muscle strength and mobility status in older adults. *Experimental Gerontology*, 91, 51–56.
- Enoka, R. M., & Duchateau, J. (2008). Muscle fatigue: What, why and how it influences muscle function. *The Journal of Physiology*, 586, 11–23.
- Garcia-Ramos, A., Gonzalez-Hernandez, J. M., Banos-Pelegrin, E., Castano-Zambudio, A., Capelo-Ramirez, F., Boullosa, D., ... Jimenez-Reyes, P. (2017). Mechanical and metabolic responses to traditional and cluster set configurations in the bench press exercise. *Journal of Strength and Conditioning Research / National Strength & Conditioning Association*. doi:10.1519/JSC.0000000000002301
- García-Ramos, A., Nebot, V., Padial, P., Valverde-Esteve, T., Pablos-Monzó, A., & Feriche, B. (2016). Effects of short inter-repetition rest periods on power output losses during the half squat exercise. *Isokinetics and Exercise Science*, 24, 323–330.
- Garcia-Ramos, A., Padial, P., Haff, G. G., Arguelles-Cienfuegos, J., Garcia-Ramos, M., Conde-Pipo, J., & Feriche, B. (2015). Effect of different interrepetition rest periods on barbell velocity loss during the ballistic bench press exercise. *Journal of Strength and Conditioning Research / National Strength & Conditioning Association*, 29, 2388–2396.
- Garcia-Ramos, A., Pestana-Melero, F. L., Perez-Castilla, A., Rojas, F. J., & Haff, G. G. (2018). Mean velocity vs. mean propulsive velocity vs. peak velocity: Which variable determines bench press relative load with higher reliability?. *Journal of Strength and Conditioning Research*, 32(5), 1273–1279.
- Girman, J. C., Jones, M. T., Matthews, T. D., & Wood, R. J. (2014). Acute effects of a cluster-set protocol on hormonal, metabolic and performance measures in resistance-trained males. *European Journal Sport Sciences*, 14, 151–159.
- Gonzalez-Badillo, J. J., Rodriguez-Rosell, D., Sanchez-Medina, L., Ribas, J., Lopez-Lopez, C., Mora-Custodio, R., ... Pareja-Blanco, F. (2016). Short-term recovery following resistance exercise leading or not to failure. *International Journal of Sports Medicine*, 37, 295–304.
- Gonzalez-Badillo, J. J., & Sanchez-Medina, L. (2010). Movement velocity as a measure of loading intensity in resistance training. *International Journal of Sports Medicine*, 31, 347–352.
- Gonzalez-Badillo, J. J., Yanez-Garcia, J. M., Mora-Custodio, R., & Rodriguez-Rosell, D. (2017). Velocity loss as a variable for monitoring resistance exercise. *International Journal of Sports Medicine*, 38, 217–225.
- González-Hernández, J. M., Garcia-Ramos, A., Capelo-Ramirez, F., Castano-Zambudio, A., Marquez, G., Boullosa, D., & Jimenez-Reyes, P. (2017). Mechanical, metabolic, and perceptual acute responses to different set configurations in full squat. *Journal of Strength and Conditioning Research*. doi:10.1519/JSC.0000000000002117
- Gorostiaga, E. M., Navarro-Amezqueta, I., Calbet, J. A., Hellsten, Y., Cusso, R., Guerrero, M., ... Izquierdo, M. (2012). Energy metabolism during repeated sets of leg press exercise leading to failure or not. *PLoS One*, 7, e40621.
- Haff, G. G., Whitley, A., McCoy, L. B., O'Bryant, H. S., Kilgore, J. L., Haff, E. E., ... Stone, M. H. (2003). Effects of different set configurations on barbell velocity and displacement during a clean pull. *Journal of Strength and Conditioning Research / National Strength & Conditioning Association*, 17, 95–103.
- Hansen, K. T., Cronin, J. B., & Newton, M. J. (2011). The effect of cluster loading on force, velocity, and power during ballistic jump squat training. *International Journal of Sports Physiology and Performance*, 6, 455–468.
- Hardee, J. P., Triplett, N. T., Utter, A. C., Zwetsloot, K. A., & McBride, J. M. (2012). Effect of interrepetition rest on power output in the power clean. *Journal of Strength and Conditioning Research / National Strength & Conditioning Association*, 26, 883–889.
- Harris, R. C., Edwards, R. H., Hultman, E., Nordesjö, L. O., Nyland, B., & Sahlin, K. (1976). The time course of phosphorylcreatine resynthesis during recovery of the quadriceps muscle in man. *Pflügers Archiv: European Journal of Physiology*, 367, 137–142.
- Hedges, L. V., & Olkin, I. O. (1985). *Statistical methods for meta-analysis*. Orlando FL: Academic Press.
- Hopkins, W. G., Marshall, S. W., Batterham, A. M., & Hanin, J. (2009). Progressive statistics for studies in sports medicine and exercise science. *Medicine and Science in Sports and Exercise*, 41, 3–13.

- Hoppeler, H. (2016). Molecular networks in skeletal muscle plasticity. *The Journal of Experimental Biology*, 219, 205–213.
- Iglesias-Soler, E., Carballeira, E., Sanchez-Otero, T., Mayo, X., Jimenez, A., & Chapman, M. L. (2012). Acute effects of distribution of rest between repetitions. *International Journal of Sports Medicine*, 33, 351–358.
- Izquierdo, M., Gonzalez-Badillo, J. J., Hakkinen, K., Ibanez, J., Kraemer, W. J., Altadill, A., ... Gorostiaga, E. M. (2006). Effect of loading on unintentional lifting velocity declines during single sets of repetitions to failure during upper and lower extremity muscle actions. *International Journal of Sports Medicine*, 27, 718–724.
- Izquierdo, M., Ibanez, J., Gonzalez-Badillo, J. J., Hakkinen, K., Ratamess, N. A., Kraemer, W. J., ... Gorostiaga, E. M. (2006). Differential effects of strength training leading to failure versus not to failure on hormonal responses, strength, and muscle power gains. *Journal Applications Physiological* (1985), 100, 1647–1656.
- Izquierdo-Gabarrén, M., Gonzalez De Txabarri Exposito, R., Garcia-Pallares, J., Sanchez-Medina, L., De Villarreal, E. S., & Izquierdo, M. (2010). Concurrent endurance and strength training not to failure optimizes performance gains. *Medica Sciences Sports Exerc*, 42(6), 1191–1199.
- Jones, K., Bishop, P., Hunter, G., & Fleisig, G. (2001, Aug). The effects of varying resistance-training loads on intermediate- and high-velocity-specific adaptations. *Journal of Strength and Conditioning Research*, 15, 349–356.
- Krieger, J. W. (2010). Single vs. multiple sets of resistance exercise for muscle hypertrophy: A meta-analysis. *Journal of Strength and Conditioning Research / National Strength & Conditioning Association*, 24, 1150–1159.
- Lawton, T. W., Cronin, J. B., & Lindsell, R. P. (2006). Effect of interrepetition rest intervals on weight training repetition power output. *Journal of Strength and Conditioning Research / National Strength & Conditioning Association*, 20, 172–176.
- Moir, G. L., Graham, B. W., Davis, S. E., Guers, J. J., & Witmer, C. A. (2013). Effect of cluster set configurations on mechanical variables during the deadlift exercise. *Journal of Human Kinetics*, 39, 15–23.
- Moreno, S. D., Brown, L. E., Coburn, J. W., & Judelson, D. A. (2014). Effect of cluster sets on plyometric jump power. *Journal of Strength and Conditioning Research / National Strength & Conditioning Association*, 28, 2424–2428.
- Oliver, J. M., Kreutzer, A., Jenke, S., Phillips, M. D., Mitchell, J. B., & Jones, M. T. (2015). Acute response to cluster sets in trained and untrained men. *European Journal of Applied Physiology*, 115, 2383–2393.
- Pareja-Blanco, F., Rodriguez-Rosell, D., Sanchez-Medina, L., Gorostiaga, E. M., & Gonzalez-Badillo, J. J. (2014). Effect of movement velocity during resistance training on neuromuscular performance. *International Journal of Sports Medicine*, 35, 916–924.
- Pareja-Blanco, F., Rodriguez-Rosell, D., Sanchez-Medina, L., Sanchis-Moysi, J., Dorado, C., Mora-Custodio, R., ... Gonzalez-Badillo, J. J. (2017). Effects of velocity loss during resistance training on athletic performance, strength gains and muscle adaptations. *Scandinavian Journal of Medicine & Science in Sports*, 27, 724–735.
- Pereira, G., Almeida, A. G., Rodacki, A. L., Ugrinowitsch, C., Fowler, N. E., & Kokubun, E. (2008). The influence of resting period length on jumping performance. *Journal of Strength and Conditioning Research / National Strength & Conditioning Association*, 22, 1259–1264.
- Pyne, D. B., Boston, T., Martin, D. T., & Logan, A. (2000). Evaluation of the Lactate Pro blood lactate analyser. *European Journal of Applied Physiology*, 82, 112–116.
- Rahimi, R., Qaderi, M., Faraji, H., & Boroujerdi, S. S. (2010). Effects of very short rest periods on hormonal responses to resistance exercise in men. *Journal of Strength and Conditioning Research / National Strength & Conditioning Association*, 24, 1851–1859.
- Sanchez-Medina, L., & Gonzalez-Badillo, J. J. (2011). Velocity loss as an indicator of neuromuscular fatigue during resistance training. *Medicine and Science in Sports and Exercise*, 43, 1725–1734.
- Sánchez-Medina, L., Pallarés, J. G., Pérez, C. E., Morán-Navarro, R., & González-Badillo, J. J. (2017). Estimation of relative load from bar velocity in the full back squat exercise. *Sports Medica International Open*, 1, E80–E88.
- Sanchez-Medina, L., Perez, C. E., & Gonzalez-Badillo, J. J. (2010). Importance of the propulsive phase in strength assessment. *International Journal of Sports Medicine*, 31, 123–129.
- Tufano, J. J., Brown, L. E., & Haff, G. G. (2017). Theoretical and practical aspects of different cluster set structures: A systematic review. *Journal of Strength and Conditioning Research / National Strength & Conditioning Association*, 31, 848–867.